2110201: COMP ENG MATH (2024/2) Quiz 3

Name: Student ID: Entry Number:

1. (4 points) Find the determinant of the matrix $\begin{bmatrix} 2 & 4 & 1 & 3 \\ 4 & 9 & 3 & 7 \\ 1 & 2 & 2 & 2 \\ 3 & 7 & 1 & -4 \end{bmatrix}$.

   [**Hint:** The answer is a two-digit positive/negative integer.]

2. (4 points) Find the inverse (if any) of the matrix $\begin{bmatrix} -1 & -2 & 0 \\ 2 & 1 & -1 \\ -1 & 0 & 1 \end{bmatrix}$.

3. (2 points) Given a matrix $\mathbf{A}_{3\times 3}$ such that $\mathbf{A}$ is a <u>diagonal</u> matrix and $\mathbf{I}$ is an identity matrix of size 3 x 3. If we know that every entry of $\mathbf{A}$ is either a <u>positive integer</u> or <u>zero</u> such that $\det(\mathbf{A}) = 8$ and $3\mathbf{A}^2 - 3\mathbf{I}$ is invertible, find the value of $\det(\mathbf{A} + \mathbf{I})$. Show your work.